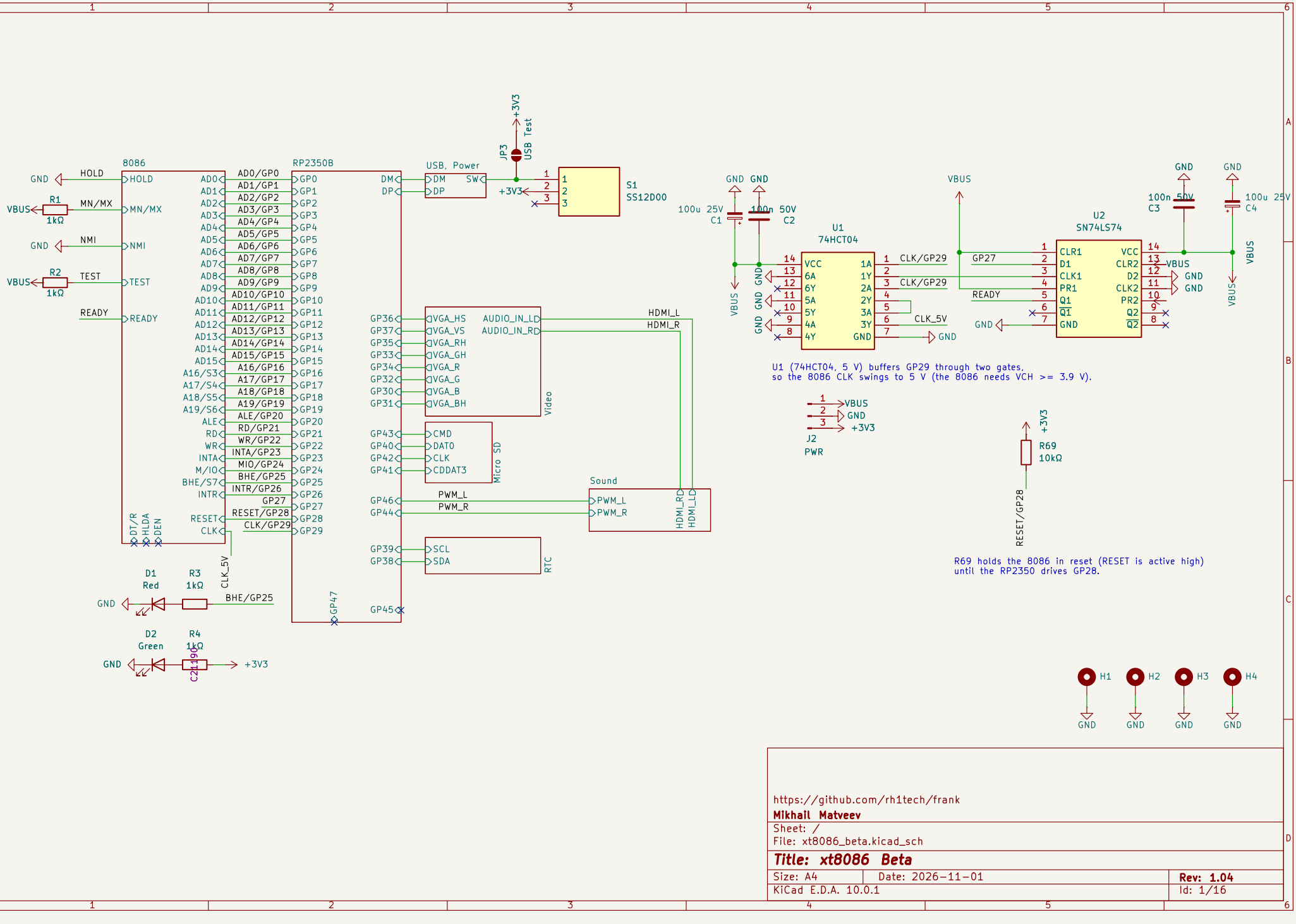


https://github.com/rh1tech/frank

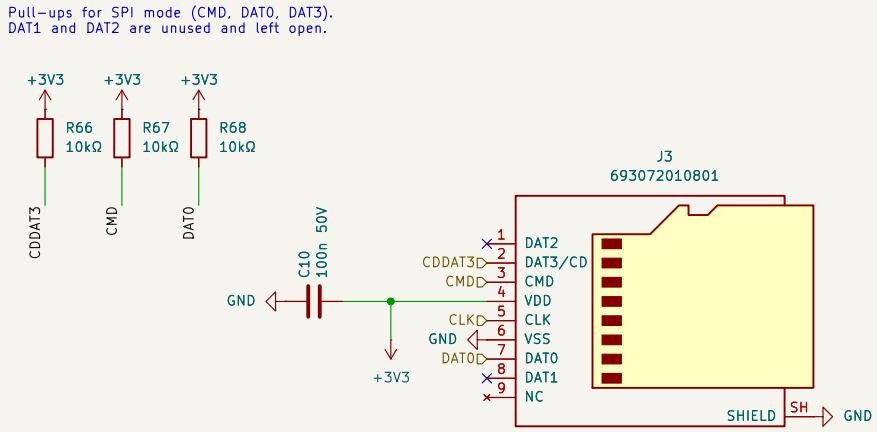
Mikhail Matveev

Sheet: /

File: xt8086_beta.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 1/16



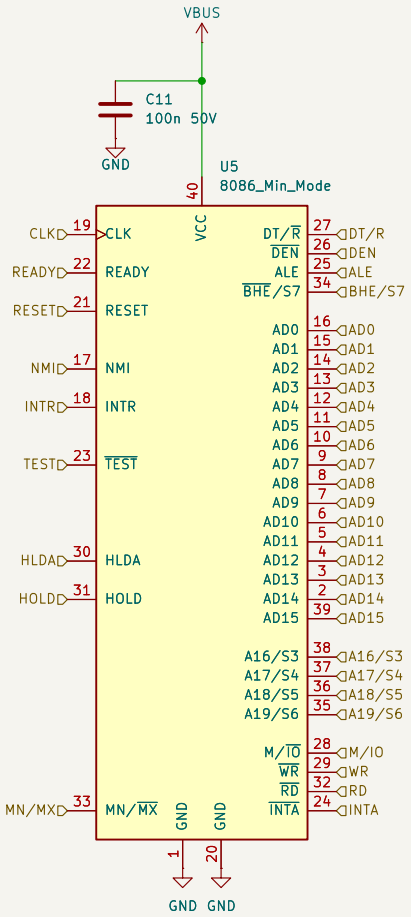
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /Micro SD/
File: sd.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 5/16



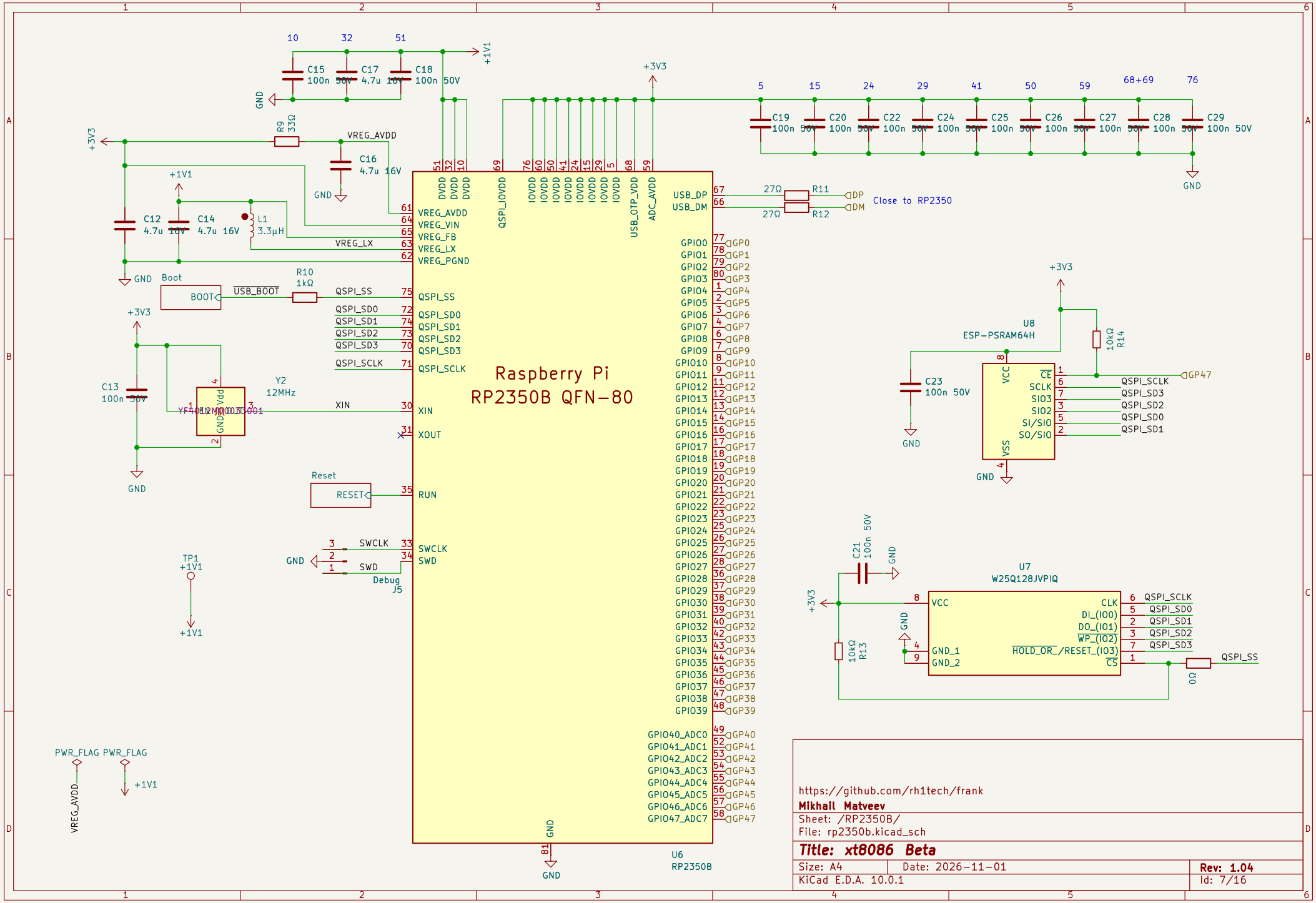
https://github.com/rh1tech/frank

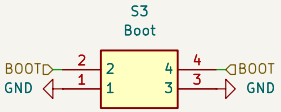
Mikhail Matveev

Sheet: /8086/
File: 8086.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 6/16


Raspberry Pi
RP2350B QFN-80
U6
RP2350B
U8
ESP-PSRAM64H
U7
W25Q128JVPIQ
Y2
12MHz
Boot
Reset
Debug
J5
TP1
Close to RP2350
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /RP2350B/
File: rp2350b.kicad_sch
Title: xt8086 Beta
Size: A4
Date: 2026-11-01
Rev: 1.04
KiCad E.D.A. 10.0.1
Id: 7/16



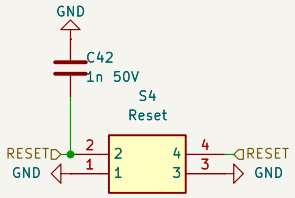
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /RP2350B/Boot/
File: boot.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 4/16



https://github.com/rh1tech/frank

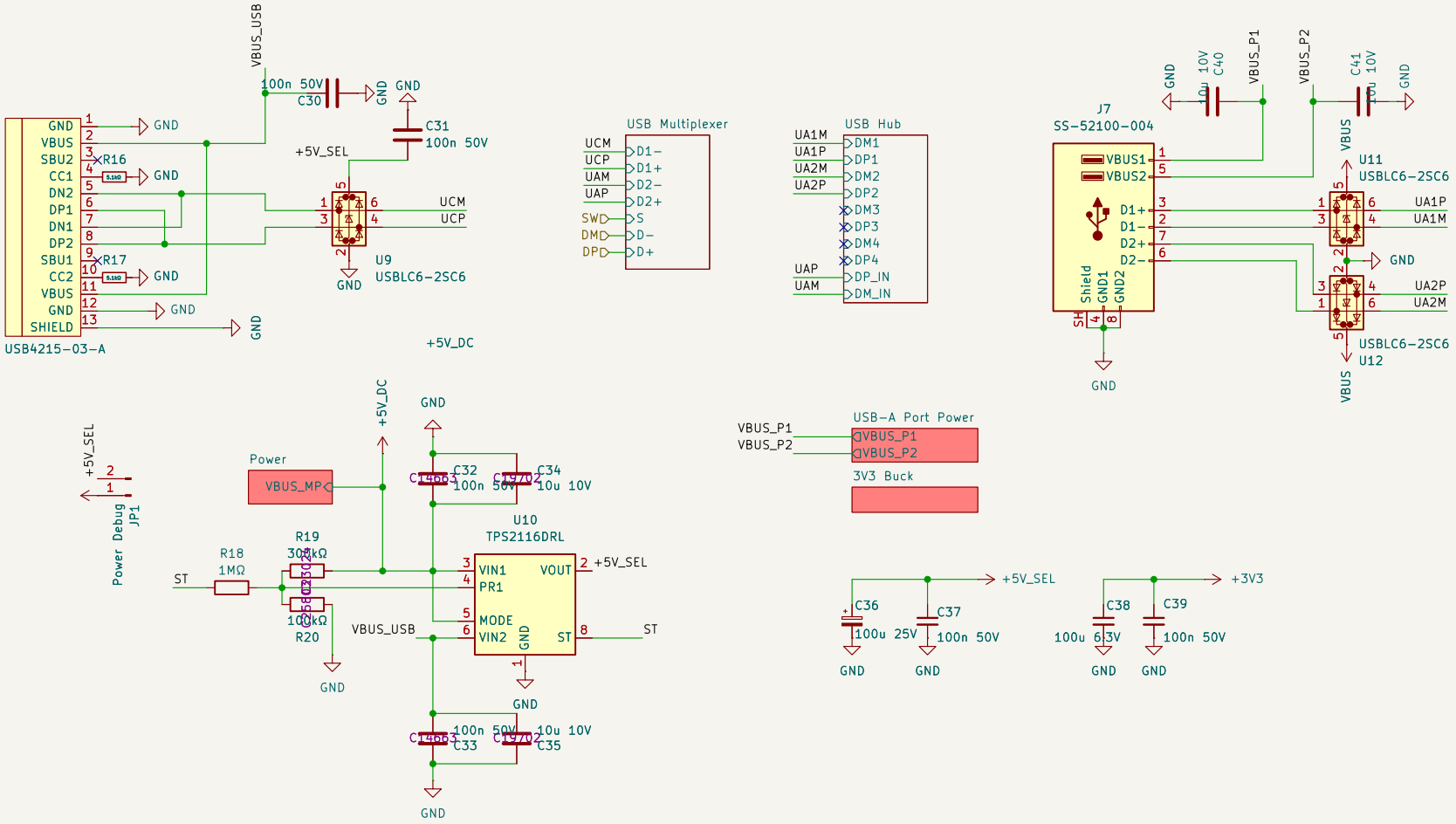
Mikhail Matveev

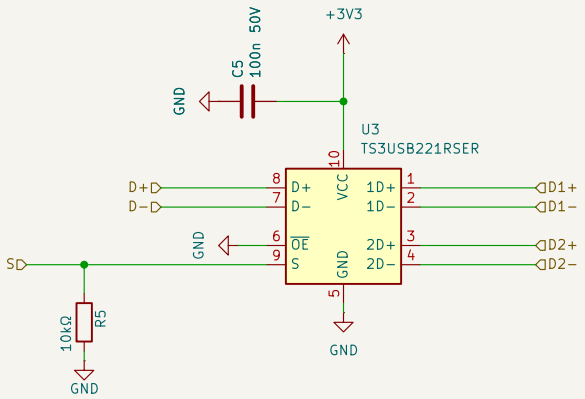
Sheet: /RP2350B/Reset/

File: reset.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 9/16


USB4215-03-A
USB Multiplexer
USB Hub
SS-52100-004
USBLC6-2SC6
USBLC6-2SC6
USB-A Port Power
3V3 Buck
Power
Power Debug
TPS2116DRL
+5V_DC
+5V_SEL
+3V3
VBUS_USB
VBUS_P1
VBUS_P2
ST
UCM
UCP
UA1P
UA1M
UA2P
UA2M
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /USB, Power/
File: usb.kicad_sch
Title: xt8086 Beta
Size: A4
Date: 2026-11-01
Rev: 1.04
KiCad E.D.A. 10.0.1
Id: 8/16



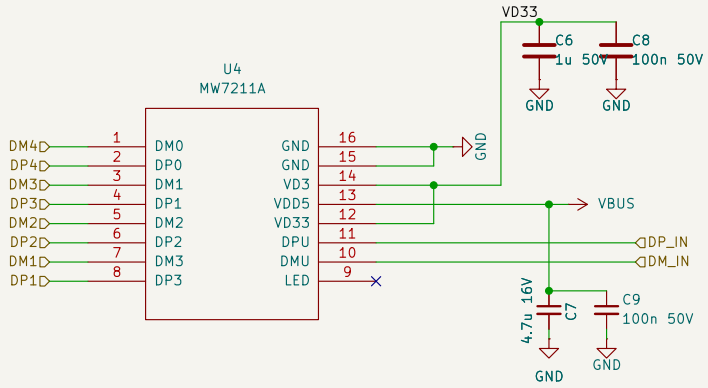
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /USB, Power/USB Multiplexer/
File: usb_multiplexer.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 2/16



https://github.com/rh1tech/frank

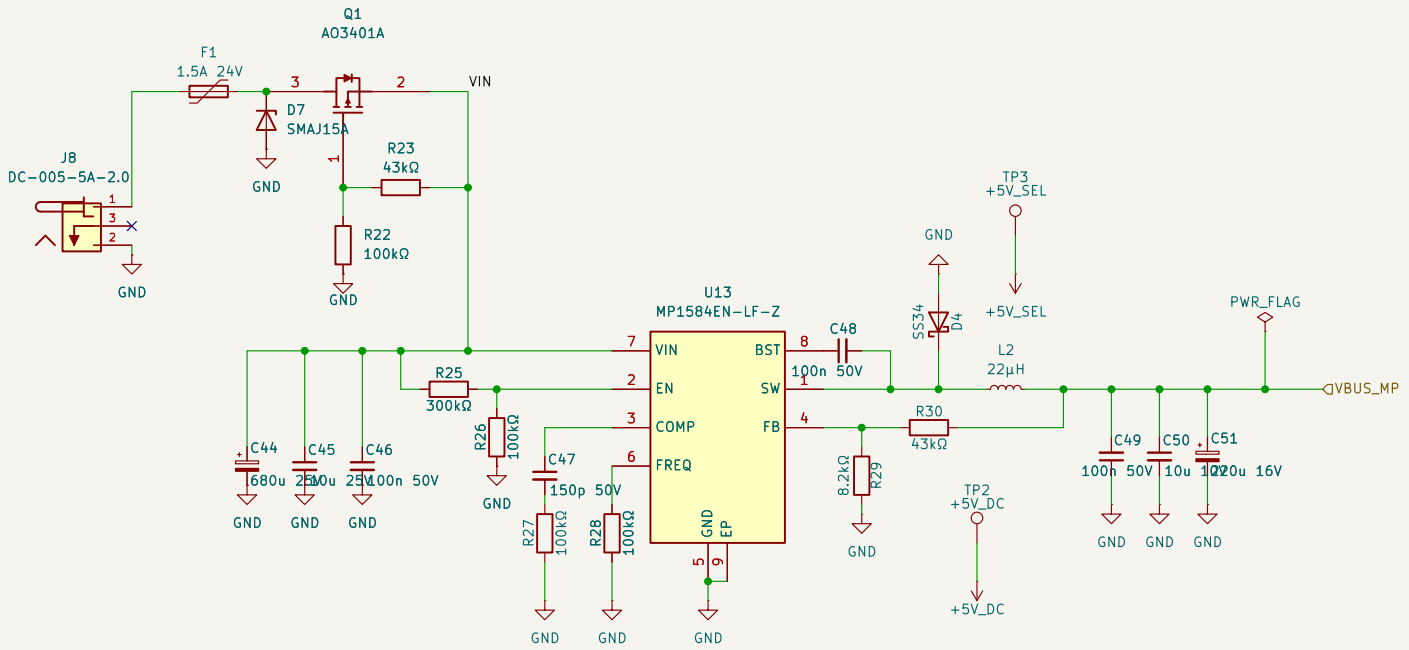
Mikhail Matveev

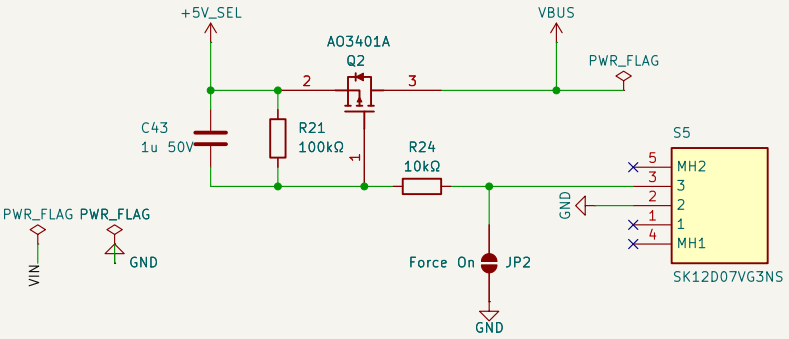
Sheet: /USB, Power/USB Hub/
File: usb_hub.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 3/16

DC input: 9–12 V (centre +), max 15 V: D7 SMAJ15A starts conducting at ~16.7 V and crowbars F1 on reverse polarity.
Q1 reverse-polarity P-FET: gate divider R22 100K / R23 43K keeps Vgs at −2.7 V (9 V) .. −7.3 V (TVS clamp).
MP1584 UVLO: R25 300K / R26 100K -> starts at ~6.0 V (6.6 V max). C44 680u 25V, C45 10u 25V.





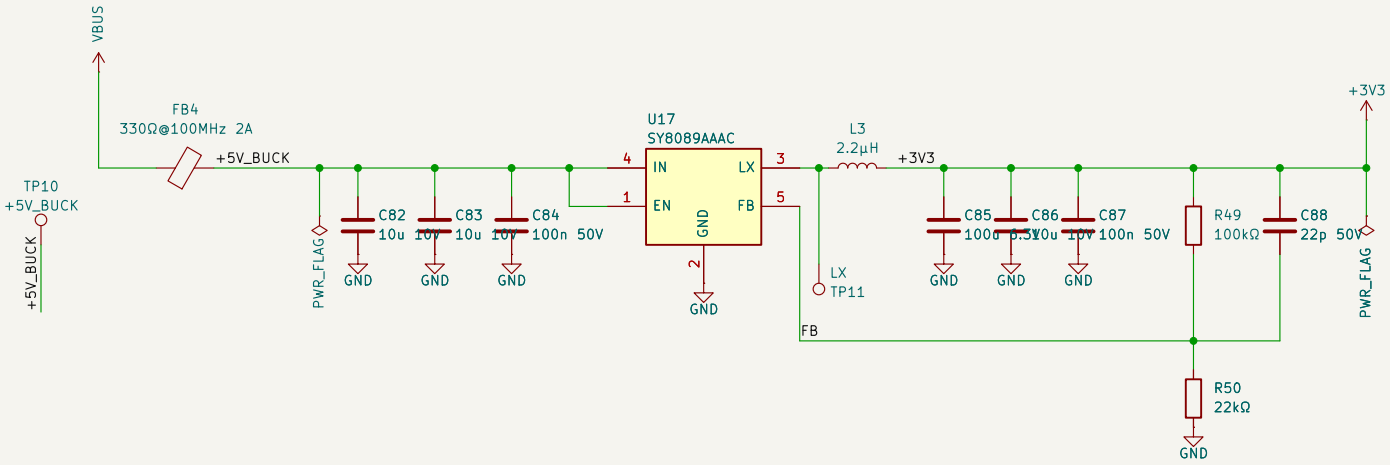
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /USB, Power/Power/
File: power.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 10/16



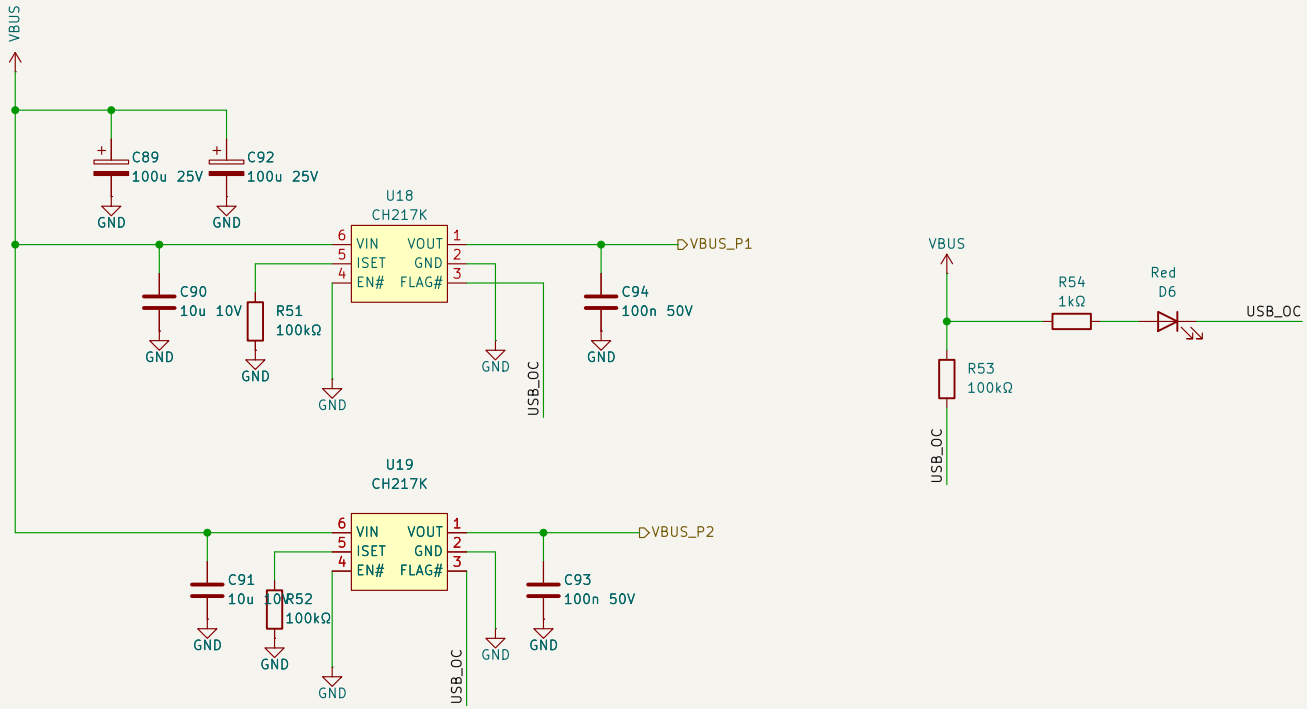
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /USB, Power/3V3 Buck/
File: buck3v3.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 17/16



https://github.com/rh1tech/frank

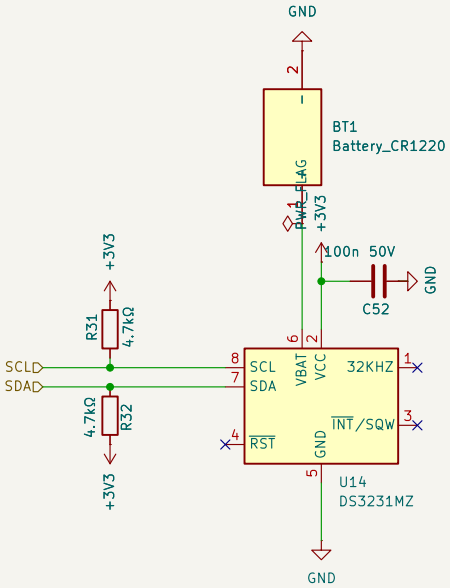
Mikhail Matveev

Sheet: /USB, Power/USB-A Port Power/

File: usb_a_power.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 18/16



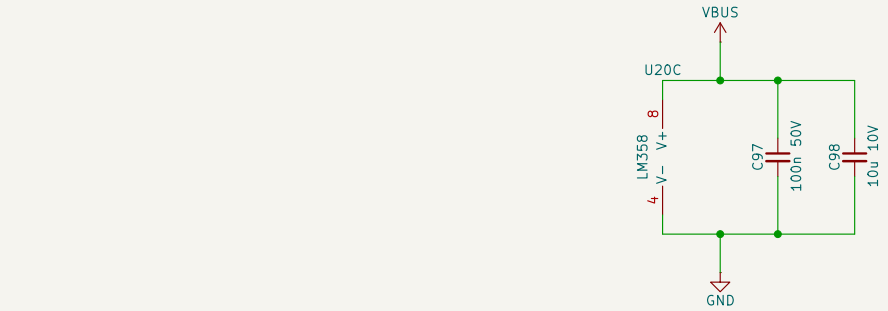
https://github.com/rh1tech/frank

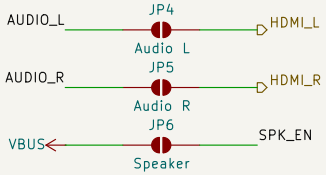
Mikhail Matveev

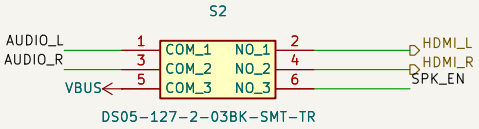
Sheet: /RTC/
File: rtc.kicad_sch

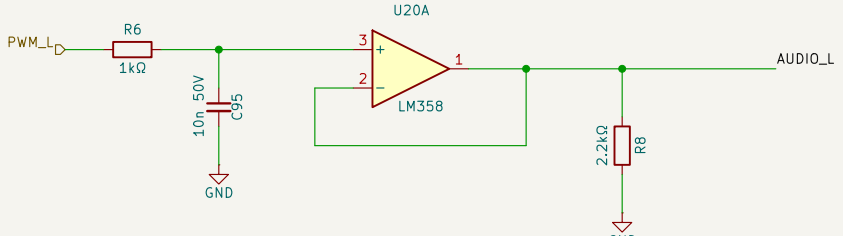
Title: xt8086 Beta

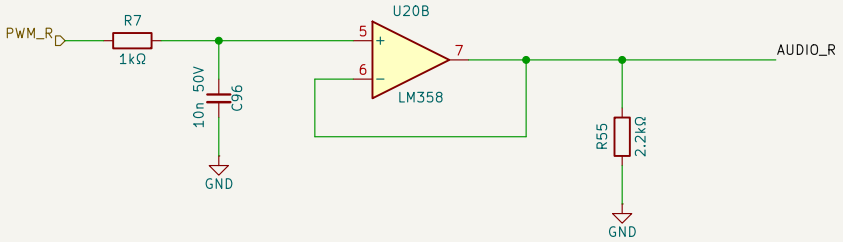
Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 11/16

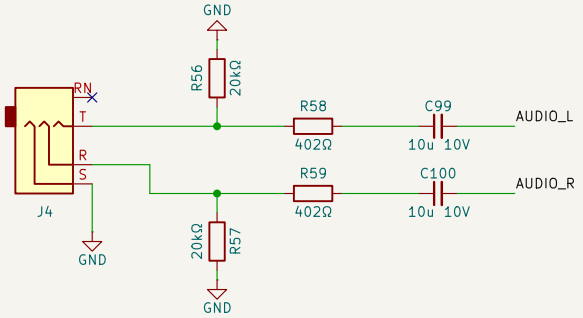


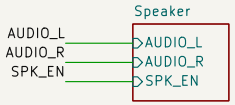












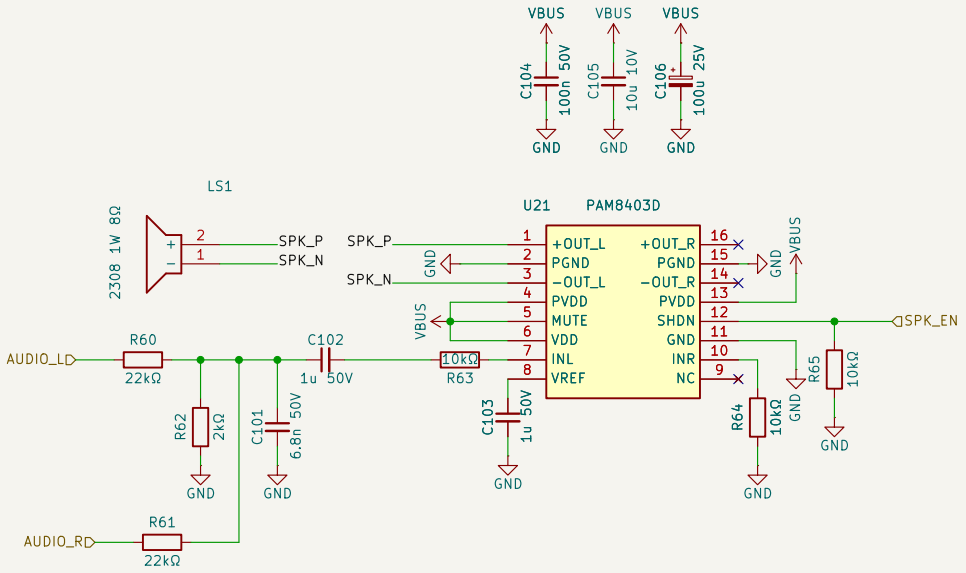
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /Sound/
File: sound.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 12/16



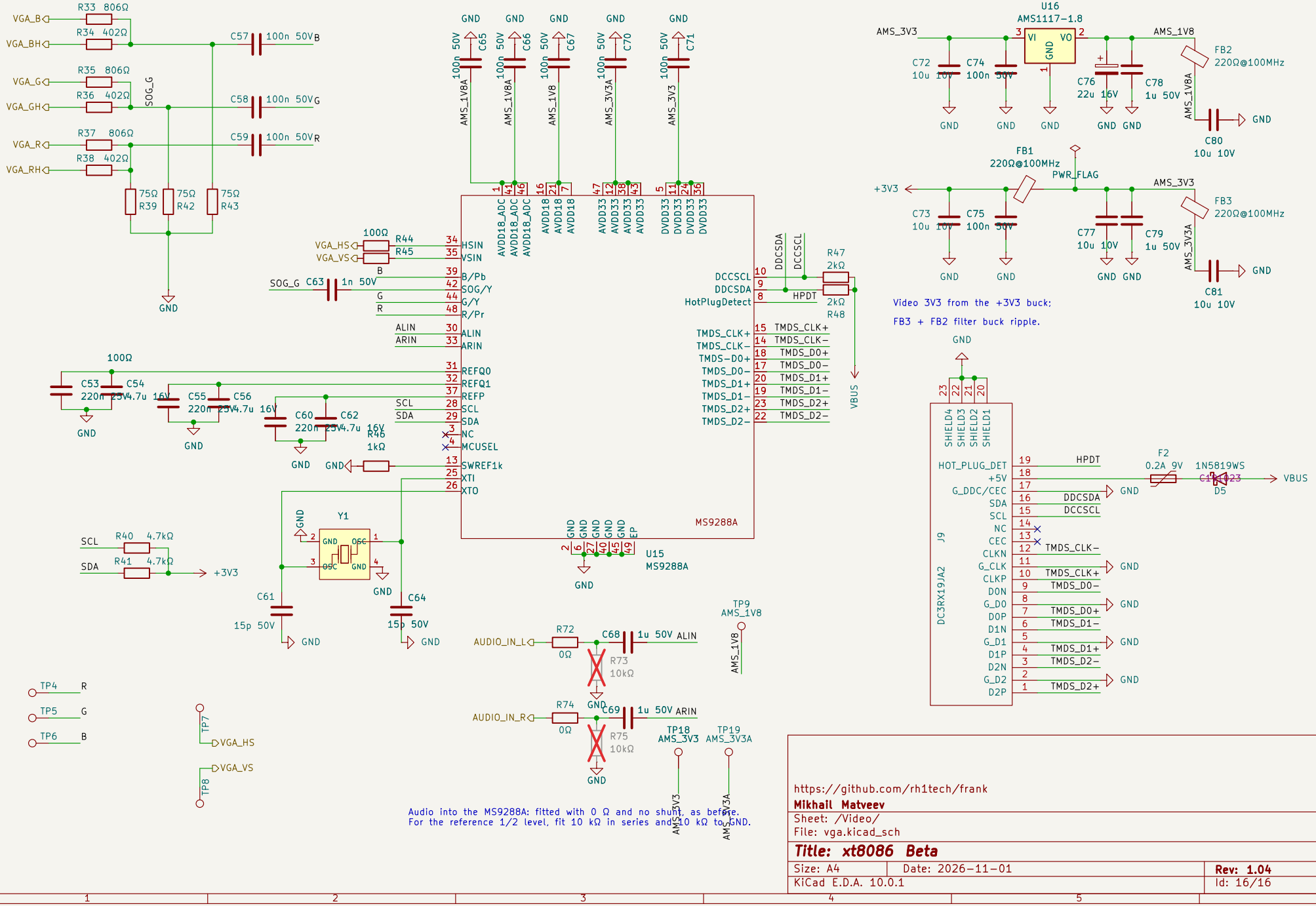
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /Sound/Speaker/
File: speaker.kicad_sch

Title: xt8086 Beta

Size: A4	Date: 2026-11-01	**Rev: 1.04**
KiCad E.D.A. 10.0.1		Id: 13/16


VGA_B
VGA_BH
VGA_G
VGA_GH
VGA_R
VGA_RH
R33 806Ω
R34 402Ω
R35 806Ω
R36 402Ω
R37 806Ω
R38 402Ω
SOG_G
C57 100n 50V B
C58 100n 50V G
C59 100n 50V R
75Ω R39
75Ω R42
75Ω R43
GND
C65 100n 50V AMS_1V8A
C66 100n 50V AMS_1V8A
C67 100n 50V AMS_1V8
C70 100n 50V AMS_3V3A
C71 100n 50V AMS_3V3
AVDD18_ADC
AVDD18_ADC
AVDD18_ADC
AVDD18
AVDD18
AVDD18
AVDD33
AVDD33
AVDD33
AVDD33
DVDD33
DVDD33
DVDD33
DVDD33
VGA_HS
VGA_VS
100Ω R44
R45
HSIN
VSIN
SOG_G C63 1n 50V
B/Pb
SOG/Y
G/Y
R/Pr
ALIN
ARIN
REFQ0
REFQ1
REFP
SCL
SDA
NC
MCUSEL
SWREF1k
XTI
XTO
C53 220n
C54 4.7u 16V
C55 220n
C56 4.7u 16V
C60 220n
C62 4.7u 16V
R46 1kΩ
GND
DCCSCL
DDCSDA
HotPlugDetect
R47 2kΩ
R48 2kΩ
HPDT
VBUS
TMDS_CLK+
TMDS_CLK−
TMDS_D0+
TMDS_D0−
TMDS_D1+
TMDS_D1−
TMDS_D2+
TMDS_D2−
MS9288A
U15
Y1
SCL
SDA
R40 4.7kΩ
R41 4.7kΩ
+3V3
C61 15p 50V
C64 15p 50V
AUDIO_IN_L
AUDIO_IN_R
R72 0Ω
R74 0Ω
C68 1u 50V ALIN
C69 1u 50V ARIN
R73 10kΩ
R75 10kΩ
TP9 AMS_1V8
TP18 AMS_3V3
TP19 AMS_3V3A
TP4 R
TP5 G
TP6 B
TP7 VGA_HS
TP8 VGA_VS
Audio into the MS9288A: fitted with 0 Ω and no shunt, as before.
For the reference 1/2 level, fit 10 kΩ in series and 10 kΩ to GND.
U16
AMS1117-1.8
AMS_3V3
AMS_1V8
C72 10u 10V
C74 100n 50V
C76 22u 16V
C78 1u 50V
FB2 220Ω@100MHz
C80 10u 10V
FB1 220Ω@100MHz
PWR_FLAG
+3V3
C73 10u 10V
C75 100n 50V
C77 10u 10V
C79 1u 50V
FB3 220Ω@100MHz
C81 10u 10V
Video 3V3 from the +3V3 buck;
FB3 + FB2 filter buck ripple.
J9
DC3RX19JA2
SHIELD4
SHIELD3
SHIELD2
SHIELD1
HOT_PLUG_DET
+5V
G_DDC/CEC
SDA
SCL
NC
CEC
CLKN
G_CLK
CLKP
D0N
G_D0
D0P
D1N
G_D1
D1P
D2N
G_D2
D2P
F2 0.2A 9V
1N5819WS
D5
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /Video/
File: vga.kicad_sch
Title: xt8086 Beta
Size: A4
Date: 2026-11-01
KiCad E.D.A. 10.0.1
Rev: 1.04
Id: 16/16